

The OpenStack compute service, primarily known as **Nova**, is the core engine responsible for managing the complete lifecycle of virtual machines (VMs) in a cloud environment 1, 2. Its architecture is designed to be highly scalable and secure through the coordination of several specialized components.

Purpose of the Compute Service

The fundamental purpose of Nova is to provide **automated virtual machine provisioning and management** 3. It acts as the "conductor" for the VM lifecycle, handling critical tasks such as:

- **Provisioning and launching** new virtual instances 1, 4.
- **Managing operational states**, including starting, stopping, rebooting, and deleting VMs 1, 5, 6.
- **Efficiently distributing resources** (processing power, memory, and storage) across physical hardware to serve end users 7, 8.

Components of the Compute Service

Nova's architecture is composed of several daemons that communicate over a **message bus** 8, 9:

- **nova-api**: The entry point for the compute service. it receives user requests via RESTful API calls and communicates them to other components over the message bus 8, 9.
- **nova-scheduler**: Known as the "**brain**" of VM placement 10. It listens for new instance requests on the message bus and selects the most suitable physical host based on resource availability and user preferences 9, 11.
- **nova-conductor**: A vital security component that acts as an **intermediary between compute nodes and the OpenStack database** 3, 12. It offloads database queries from the compute nodes to prevent a compromised host from gaining direct access to the central database 12, 13.
- **nova-compute**: The service running on the **physical compute nodes** 12. It is responsible for the actual starting and terminating of virtual machines 9. Notably, it lacks direct access to the database to minimize security risks 9, 14.

Interaction with Other OpenStack Modules

Nova does not function in isolation; it orchestrates resources by interacting with the following modules:

- **Keystone (Identity Service)**: Handles authentication and authorization; users must provide a token from Keystone to interact with the Nova API 15, 16.
- **Glance (Image Service)**: Acts as the repository for **VM images (templates)** 5, 17. Nova retrieves these templates to deploy new instances 4, 18.
- **Neutron (Networking Service)**: Provisions the virtual networks, subnets, and IP addresses required for VM connectivity 4, 5, 19.
- **Cinder (Block Storage Service)**: Provides persistent block storage volumes that function as virtual hard disks for the VMs 5, 18, 20.

- **Telemetry (Ceilometer):** Collects data on VM performance and resource utilization for monitoring and troubleshooting 21-23.

Instance Lifecycle Handling

The process of launching and managing an instance involves several steps:

1. **Authentication & Request:** A user submits a request via the **Nova API** or **Horizon Dashboard**, which is authenticated by **Keystone** 4, 6, 16.
2. **Scheduling:** The **nova-scheduler** evaluates the request against available physical hosts using a process of **filtering** (removing unsuitable nodes) and **weighting** (ranking suitable nodes) to find the optimal host 9, 24, 25.
3. **Resource Gathering:** Once a host is selected, the service requests the VM template from **Glance** and networking configurations from **Neutron** 4.
4. **Deployment:** The **nova-compute** service on the selected physical node receives the request via the message bus and launches the VM 9.
5. **Ongoing Management:** Throughout its life, the VM's state (start, stop, etc.) is managed via the API, and its resource usage is tracked by **telemetry services** 1, 21, 22.

Exam-Ready Explanation

Question: Describe the Architecture and Functionality of OpenStack Nova.

- **Core Role:** Nova is the **Compute Service** responsible for the automated lifecycle management of virtual machines 1, 2.
- **Key Security Principle: Database Isolation.** Compute nodes (nova-compute) are denied direct database access. Instead, they must use the nova-conductor as a proxy to protect the central database from potentially compromised hosts 9, 12, 14.
- **The Scheduling Logic:** The nova-scheduler uses a three-step process: **Filter** (remove invalid hosts), **Weight** (rank valid hosts), and **Select** (choose the host with the highest weight) 24, 25.
- **Horizontal Scalability:** Components like the nova-conductor can be deployed across multiple machines to handle growing workloads 13.
- **Service Integration:**
 - **Glance** = Templates 18.
 - **Neutron** = Connectivity 19.
 - **Cinder** = Persistent Disks 18.
 - **Keystone** = Security/Tokens 16.
- **Management Tip:** For maximum security, **never co-locate** nova-conductor and nova-compute on the same physical machine 10, 13.